

First Terminal Examination - 2080

Class : XI

Subject: Biology (Code: 2011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet. [6 × 1 = 6]

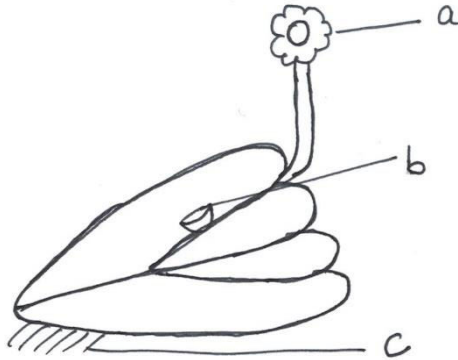
1. Which one of the following is cytoplasmic ribosome in a eukaryotic cell?
 - a. 80s
 - b. 70s
 - c. 50s
 - d. 60s
2. Entry of liquid substance in the cell is called
 - a. Exocytosis
 - b. Pinocytosis
 - c. Endocytosis
 - d. Phagocytosis
3. Which one of the following is not present in eukaryotes?
 - a. Ribosomes
 - b. Mitochondria
 - c. Incipient nucleus
 - d. Centrioles
4. Ladder like structure in *Spirogyra* is seen during
 - a. Conjugation
 - b. Sexual reproduction
 - c. Homothallic conjugation
 - d. Heterothallic conjugation
5. The smallest unit of hierarchic system of classification is
 - a. Genus
 - b. Species
 - c. Order
 - d. Class
6. are photosynthetic, non-vascular, thalloid aquatic plants
 - a. Bryophytes
 - b. Pteridophytes
 - c. Algae
 - d. Gymnosperms

Group 'B' [BOTANY]

Give short answer to the following questions.

[4 × 4 = 16]

1. Distinguish between prokaryotic and eukaryotic cells. [4]
2. Study the diagram given below & answer the questions. [4×1 = 4]



- a) What plant is shown in the diagram?
 - b) Name the labelled parts a, b & c.
3. Name the structure which is:
 - a) Physical basis of life
 - b) Cytoplasmic bridge between the adjacent cells in plants
 - c) Suicidal bag of a cell
 - d) Protein factory of the cell
 4. Why bryophytes are called amphibians of plant kingdom? Distinguish between male & female plants of *Marchantia* with necessary sketches. [1+3]

OR

Define binomial system of nomenclature. Write down its rules. [1+3]

Group 'C' [BOTANY]

Give long answer to the following questions.

[8 × 2 = 16]

5. Name the starch storing Leucoplast. Why chloroplast is called semi-autonomous cell organelle? Describe the structure & functions of chloroplast. [1+1+3+3]
6. What do you mean by thallus? What is the function of mucilage in *Spirogyra*? Describe conjugation process in *Spirogyra* in detail. [1+1+6]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet. [5 × 1 = 5]

1. Study of tissue/microscopic anatomy is termed as
 - a) Ecology
 - b) Cytology
 - c) Histology
 - d) Anatomy
2. Paleontology is the study of
 - a) Breasts
 - b) Birds
 - c) Fossils
 - d) Bones
3. The organs having similar function and different in structure or origin:
 - a) Homologous
 - b) Analogous
 - c) Vestigial
 - d) Parallel
4. Which of the following is connecting link between animals and plant kingdom?
 - a) *Euglena*
 - b) *Amoeba*
 - c) Dinoflagellates
 - d) Diatoms
5. The inheritance of acquired character is given by the concept of:
 - a) Darwinism
 - b) Lamarckism
 - c) Mendelism
 - d) Neo Lamarckism

Group 'B' [ZOOLOGY]

Give short answers to the following questions.

[4 × 4 = 16]

1. Briefly discuss the scopes of biology in this modern day. [2+2]
2. "Biology is the science of exception". Elaborate this statement. [2+2]
3. Explain the meaning of "Ontogeny recapitulates phylogeny". [1+3]
4. Summarize Miller's simulation experiment for organic synthesis with figure. Comment on its efficacy. [2+2]

Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

5. Name the theory of evolution purposed by Darwin. Enumerate the key concepts in the evolution theory of Darwin. [1+7]
6. Explain how life originated in earth according to Oparin and Haldane theory. [3+3+2]

❧ The End ❧